

AEM-X100 Wafer Handling System — Equipment Overview

Equipment: AEM-X100 Wafer Handling System

Document ID: SYN-X100-EQUIPMENT_OVERVIEW **Revision:** A **Status:** Synthetic Training Data

IMPORTANT: This document is entirely fictional and created for the AAS Practice Module prototype. It is not an AEM document, does not represent an actual AEM product or maintenance procedure, and must not be used for real equipment servicing.

Purpose: Provide a controlled vocabulary and high-level description of the fictional equipment used by the service-assistant prototype.

1. System Description

- The AEM-X100 is a fictional automated wafer handling system used in this prototype to move semiconductor wafers between an input load port, alignment station, inspection position, and output load port.
- The system is represented as a modular machine with motion control, wafer sensing, vacuum handling, interlock monitoring, and a supervisory control application.

1.1 Main Modules

- Load Port Module (LPM): accepts wafer carriers and presents wafers to the transfer mechanism.
- Wafer Transfer Module (WTM): moves wafers between defined stations using a robotic end effector.
- Wafer Presence Sensor (WPS): detects whether a wafer is present at designated sensing positions.
- Alignment Module (ALM): performs a fictional pre-transfer alignment check.
- Vacuum Handling Module (VHM): provides the fictional vacuum interface used by the end effector.
- Safety Interlock Module (SIM): monitors doors, covers, and emergency-stop conditions.
- Supervisory Control Unit (SCU): provides status, alarms, recipes, counters, and maintenance messages.

2. Key Terminology

Term	Meaning
LPM	Load Port Module
WTM	Wafer Transfer Module
WPS	Wafer Presence Sensor
ALM	Alignment Module
VHM	Vacuum Handling Module
SIM	Safety Interlock Module
SCU	Supervisory Control Unit
E101	Synthetic wafer-presence detection fault
E201	Synthetic transfer-position fault
E301	Synthetic safety-interlock fault

3. Normal Operating States

- IDLE — system available and no active transfer.
- READY — preconditions satisfied and transfer sequence may be started.
- TRANSFER — wafer movement sequence is active.

- PAUSED — operation is temporarily stopped pending an operator action.
- FAULT — an alarm requires troubleshooting or escalation.
- SAFE STOP — operation is inhibited because a safety condition is active.

4. Information Used by the Service Assistant

- The assistant should use equipment module names, alarm codes, operating states, and documented procedures as retrieval anchors.
- When information is not available in the synthetic knowledge base, the assistant should state that the evidence is insufficient rather than inventing a procedure.
- Cross-references to the other synthetic documents are intentional and are used to test multi-document retrieval.

End of synthetic document.